Multimedia Signal processing

Prof. Marco Tagliasacchi

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1. **Question (8 pts)**:
   Consider a base station that records the number of users connected to it at discrete intervals. For example, the following sequence of symbols is observed

$$x = [10, 12, 20, 15, 30, 35, 40, 46] \tag{1}$$

   (a) Encode the sequence with PCM, using the following quantizer:

$$\begin{aligned} I_1 &= [0, 10), & y_1 &= 0 \\ I_2 &= [10, 20), & y_2 &= 10 \\ I_3 &= [20, 30), & y_3 &= 20 \\ I_4 &= [30, 40), & y_4 &= 30 \\ I_5 &= [40, +\infty), & y_5 &= 40 \end{aligned} \tag{2}$$

   Compute the MSE and the rate. For the latter, use an approximation that adopts Shannon's coding with the following pmf.

y_k	Pr.
0	0.3
10	0.3
20	0.2
30	0.1
40	0.1

   (b) Encode the sequence with DPCM, using the following quantizer (hint: use PCM for the first sample with the quantizer in (a)):

$$\begin{aligned} I_1 &= (-\infty, -15), & e_1 &= -20 \\ I_2 &= [-15, -5), & e_2 &= -10 \\ I_3 &= [-5, 5), & e_3 &= 0 \\ I_4 &= [5, 15), & e_4 &= +10 \\ I_5 &= [15, +\infty), & e_5 &= +20 \end{aligned} \tag{3}$$

   Compute the MSE and the rate. For the latter, use an approximation that adopts Shannon's coding with the following pmf.

e_k	Pr.
-20	0.01
-10	0.04
0	0.9
+10	0.04
+20	0.01

2. **Question (5 pts)**
   (a) Illustrate the block diagram of a lossy image coder, briefly describing the role of each module.
   (b) Describe how to encode color images.
3. **Question (5 pts)**
   Discuss the design of the perceptual model in audio coding.
4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Let X denote a discrete source with M symbols.
   - [T] [F] A uniquely decodable code always uses less than $\log_2 M$ bits / symbols
   - [T] [F] If $H_N(X) = H(X)$, $\forall N$, then X is a source without memory.
   - [T] [F] Any instantaneous code is also uniquely decodable.
   - [T] [F] $H(X, X) = 2H(X)$.

   (b) Consider a speech coder
   - [T] [F] In an open loop vocoder, the encoder embeds the decoder.
   - [T] [F] The long term predictor accounts for the *pitch structure*.
   - [T] [F] Let $A_L(z)$ denote the long term predictor. Then, if the filter $1/(1 - A_L(z))$ is fed with white noise, the output is periodic.

   (c) Consider a DPCM coder.
   - [T] [F] If $\Delta_1 < \Delta_2$, then $D(\Delta_1) < D(\Delta_2)$, where $D(\Delta)$ is the MSE distortion obtained with a uniform quantizer with step size Δ
   - [T] [F] The coding efficiency of DPCM is always better than, or equal to, that of PCM on any signal.
   - [T] [F] The predictor used in DPCM is constrained to be a linear predictor.
5. **Question (Matlab - 10pts)**
   Load the files *f1.mat* and *f2.mat* which contain two consecutive frames of a video sequence. Apply the following to the difference between the two frames.
   (a) Divide in 8×8 blocks and compute the 2D DCT of each block.
   (b) Plot the PSNR-Δ function obtained by quantizing the DCT coefficients, with $\Delta \in \{1, 2, 4, 8, 16\}$
   (c) Divide in 8×8 blocks and compute the 2D Hadamard transform of each block. The 2D Hadamard transform is a separable transform that can be obtained from the 1D Hadamard transform (hint: the 1D Hadamard transform matrix is obtained by means of *normc(hadamard(8))*).
   (d) Plot the PSNR-Δ function obtained by quantizing the Hadamard coefficients, with $\Delta \in \{1, 2, 4, 8, 16\}$